Performance & Responsiveness Roadmap (Long Version)

Objective

- Treat the lobby/home as a real-time system rather than a simple app.
- Ensure fast, stable, and deterministic transitions across key user journeys.
- Shift from reactive debugging to measurable, system-level optimization.

Critical User Paths

- Boot → Home ready (time-to-first-frame, stability during startup).
- Home → App launch (latency, reliability, lifecycle transitions).
- App → Home return (teardown cost, UI rehydration, shell stability).

Phase 1: Observability Baseline

- Instrument critical paths end-to-end.
- Measure latency, frame drops, crashes/ANRs, and resource spikes.
- Build a baseline dashboard with P50/P90/P99 metrics.

Phase 2: Systemic Bottleneck Identification

- Detect main-thread blocking in lobby/UI layer.
- Identify contention between 3D rendering and 2D UI composition.
- Analyze lifecycle instability during app transitions.
- Evaluate update/version mismatch risks from Play Store.

Phase 3: High-ROI Structural Fixes

- Isolate 3D shell and 2D lobby processes to reduce blast radius.
- Introduce transactional app launches with timeouts and fallback behavior.
- Add crash-loop detection and quarantine mechanisms.
- Implement update compatibility gating and version policies.

Phase 4: Targeted Performance Optimization

- Move heavy work off the main thread and adopt lazy initialization.
- Optimize UI surface updates and texture uploads.

- Tune asset loading, caching, and memory lifecycle.
- Refine 3D rendering pipeline and frame pacing.

Phase 5: Continuous Performance Governance

- Establish performance budgets for critical transitions.
- Adopt staged update rollouts and cohort-based releases.
- Automate regression detection and telemetry-driven alerts.
- Use performance KPIs as release gates.

Expected Outcomes

- Consistent and predictable home experience.
- High app launch success rate (>99.9%).
- Near-zero update-induced instability.
- Sustainable performance across device lifetime.